

Mapping & Analyzing Participation

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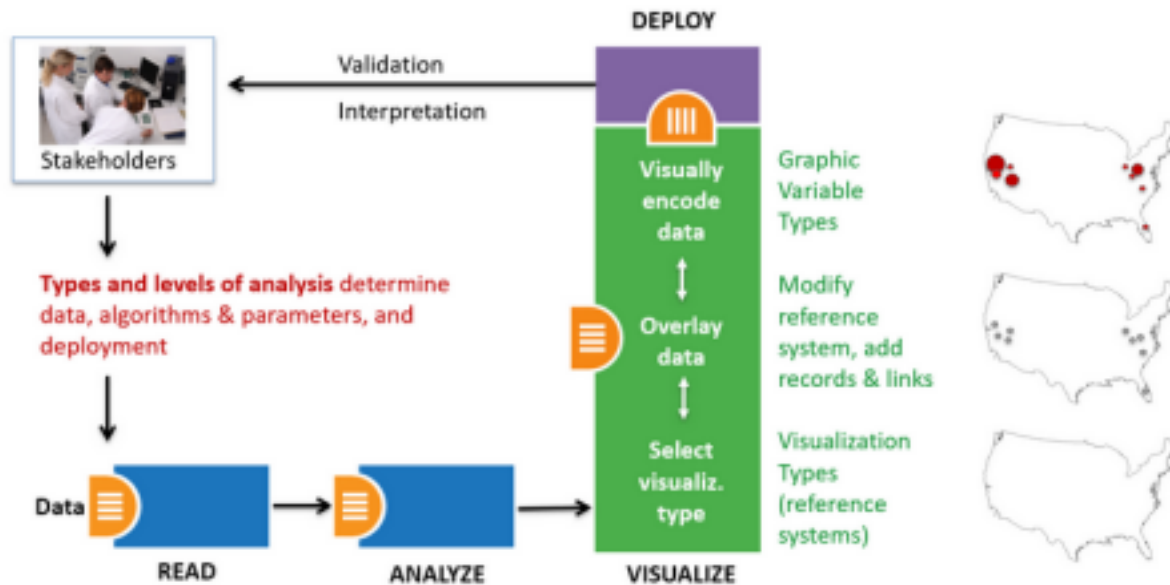


Fig. 1. The visual abstract figure should show a visually engaging summary of the project—including the problem, data used, methods/algs/tools applied, and key visualizations. Use simple graphics and minimal text. Examples are at [IEEE VIS 2024 Content: Papers](#)

1. INTRODUCTION AND PRIOR WORK

Many events bring together people from different organizations and backgrounds, but organizers often lack tools that make participation data easy to explore and interpret. This project proposes an interactive dashboard that can be built from simple, open-source methods and integrated into existing registration processes. By collecting basic information such as names, roles, organizations, and ZIP codes, the dashboard transforms raw records into clear, flexible visualizations. Instead of static tables or manual reports, stakeholders can use dropdown filters to view participation by geography, track changes over time, and distinguish between new and recurring members. Prior efforts to analyze participation often relied on manual data entry or inconsistent formats, which limited their usefulness. Our dashboard builds on that work by offering a streamlined, reproducible way to visualize engagement patterns, helping communities identify gaps, strengthen retention, and plan more inclusive events.

1.1 Stakeholder Groups

This project involves event organizers, data analysts, community leaders, and participants. Organizers want a simple way to collect and manage registration data, while analysts need clean, structured inputs for network mapping tools. Community leaders and participants benefit from maps that show who's involved and help build stronger, more inclusive networks.

1.2 Stakeholder Needs

The interactive dashboard we created is designed to meet the diverse needs of organizers, analysts, and community leaders

by making participation data both accessible and actionable. For organizers, the dashboard offers a simple interface that integrates seamlessly into their existing workflow. With dropdown filters and clear color-coded maps, they can quickly track participation by ZIP code, monitor changes over time, and distinguish between new and recurring members without needing advanced technical skills. Analysts benefit from the structured format of the dashboard, which presents multiple metrics in a consistent and interactive way. This allows them to dig deeper into trends, export data for further analysis, and ensure compatibility with tools like Power BI or other visualization platforms. Community leaders, meanwhile, gain clear visuals and summaries that highlight geographic gaps and opportunities for outreach. By toggling between views of total participation, change in participation, new members, and recurring members, they can identify underserved regions, evaluate retention, and guide future planning efforts. In this way, the dashboard serves as a shared resource that empowers each stakeholder group to explore the data in ways that directly support their goals.

2. DATA ACQUISITION

This project uses two types of data: primary data collected directly from participants through registration forms, and secondary data from public spreadsheets that track attendance at Tutor/Mentor conferences from 1994 to 2015.

2.1 Data Sources

. Primary data includes participant information collected directly through registration forms, while secondary data comes from publicly available spreadsheets documenting

attendance at the conferences. Before building the dashboard, we cleaned and standardized participant records to ensure consistency. Key fields such as ZIP code, year, and organization type were reformatted, duplicates removed, and missing values addressed. We also created derived metrics total participation, change in participation, new members, and recurring members so the dashboard could display flexible views of engagement across time and geography.

2.2 Data Description, Quality and Coverage

The dataset used in this project consists of Excel spreadsheets documenting participant information from 42 Tutor/Mentor Leadership and Networking Conferences held between May 1994 and May 2015. Each row in the dataset represents a unique participant at a specific event and includes fields such as name, organization, role/title, location, and event date. Additional categorical variables include sector, stakeholder type, and prior participation history.

The data spans over two decades and covers events primarily held in Chicago. While the dataset is comprehensive in scope, some records vary in completeness due to differences in how data was collected across years. To improve quality, the data has been cleaned for duplicates, standardized across fields, and annotated with metadata to support traceability and future reuse.

3. DATA ANALYSIS

For this stage, we worked mainly with the cleaned participation dataset covering more than twenty years of conference attendance. After merging and standardizing all available records, we focused on counting unique participants, identifying repeat attendees, and studying their distribution across states and organizations.

The data included thousands of individual entries. Most participants came from Illinois, with a strong cluster centered in Chicago and nearby cities such as Evanston, Oak Park, and Cicero. Outside Illinois, several states, including Indiana, Michigan, and Wisconsin, showed consistent participation across multiple years.

We used simple descriptive statistics to summarize this information. The total participant count increased gradually from the 1990s through the early 2010s, suggesting a steady growth in engagement. However, the participation rate dropped slightly in the final years, likely due to changes in outreach or available data coverage.

Geographic concentration: Participation spanned 616 ZIP codes across 45 states, but Illinois dominated with about 66% of all participants. Chicago alone accounted for over 5,500 entries, underscoring its role as the central hub. After 2005, participation gradually expanded into nearby states such as Indiana, Michigan, and Wisconsin, reflecting regional growth.

Organizational diversity: More than 1,532 unique organizations were represented. Nonprofit and education-related groups contributed the majority of attendees, while donor and policy-focused

organizations made up a smaller share, highlighting a sector imbalance.

Temporal trends: Yearly participation fluctuated, peaking in the late 1990s and early 2000s, followed by a gradual decline after 2010. This suggests that outreach strategies or conference accessibility may have shifted in later years.

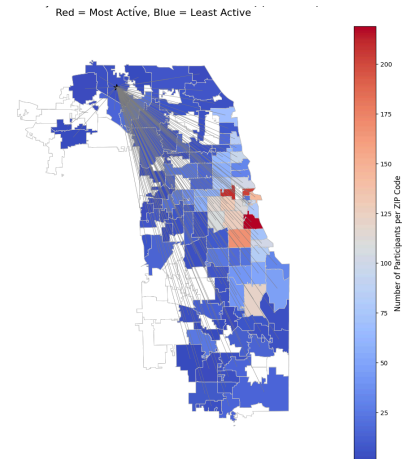


Fig. 2. This map is an initial visualization of a subsection of Illinois broken up by zip code. The zip codes are color coded with red being the most active, and blue being the least active. This was our inspiration for what our final dashboard would look like.

4. VISUALIZATIONS

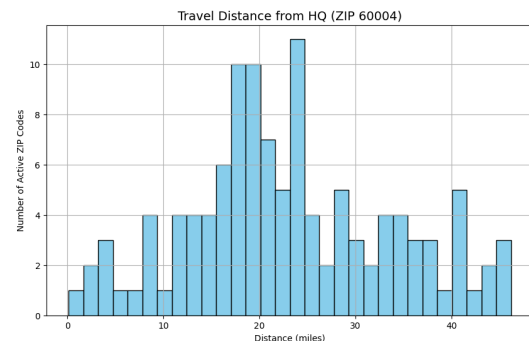


Fig. 3. The distribution of zip codes by distance to the conference center.

To look into this further, Fig 3 shows an initial histogram of the number of zip codes that fall into a specific distance bin within some amount of miles from the Tutor Mentor institute. This shows us that the average participation is roughly gaussian, and could benefit from moving the conference to a more dense region which could potentially lead to higher turnover.

Dashboard Walkthrough

In addition to static figures, an interactive dashboard was developed using Plotly to provide a more flexible view of participation trends. The dashboard displays ZIP codes shaded by intensity, where darker colors represent higher levels of activity. Users can explore the data dynamically through a dropdown menu that switches between four key metrics:

Total Participation:

Shows the overall number of participants from each ZIP code. This view highlights the most active regions and provides a baseline for geographic engagement.

Change in Participation:

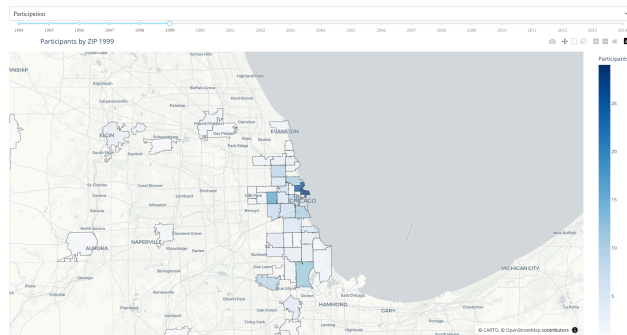
Displays year-to-year changes in participation by ZIP code. Areas shaded in stronger colors indicate growth, while lighter or negative values highlight declines. This view is useful for spotting emerging regions or places where engagement has tapered off.

New Members:

Focuses on participants attending for the first time. This metric helps identify ZIP codes that are contributing fresh engagement and expanding the community base.

Recurring Members:

Highlights ZIP codes with participants who attended multiple times.



Here is a link to the dashboard below:

<https://2a1bd5ae-6151-493c-b8d0-9a07aef7f5b0.plotly.app>

5. USAGE AND CRITIQUE OF AI TOOLS

Our group used Ai for brainstorming possible ideas of what could be beneficial for our project. There were times where the Ai couldn't understand the scale of our dataset or the categorical.

6. INTERPRETATION OF RESULTS

Our analysis has shown strong participation from nonprofits and educators, but limited involvement from donors and policymakers. This imbalance suggests that the conferences

primarily served as spaces for practitioners and community-based organizations rather than funding institutions. While this helped strengthen grassroots connections, it also limited opportunities for broader institutional influence and resource mobilization.

The interactive dashboard provides a clearer picture of how participation evolved across ZIP codes and years. By toggling between metrics such as total participation, change in participation, new members, and recurring members, several patterns emerge:

- **Community clusters:** The dashboard shows that nonprofit and education-related ZIP codes consistently appear with high intensity, especially in Illinois. Many of these areas also show strong recurring member counts, reinforcing the persistence of local and regional communities over time.
- **Peripheral actors:** When switching to the “new members” view, donor and policy-related ZIP codes appear less frequently and with weaker intensity. Their limited presence suggests missed opportunities to expand engagement beyond practitioner and educator communities.
- **Regional hubs:** Chicago and surrounding Illinois ZIP codes dominate across all dashboard views, acting as central hubs of participation. After 2005, the dashboard highlights new activity in Indiana, Michigan, and Wisconsin, showing geographic expansion and gradual integration of neighboring states into the conference network.
- **Retention vs. outreach:** The dropdown comparison between recurring members and new members reveals a balance between retention and outreach. Some ZIP codes show strong recurring participation, indicating long-term commitment, while others contribute more new members, highlighting successful outreach into fresh regions.

Together, these dashboard insights suggest that while the conferences were highly effective at sustaining nonprofit and education networks, future outreach could focus on strengthening donor and policy participation and leveraging regional growth beyond Illinois.

ACKNOWLEDGEMENTS

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